A company works with number of employees; all the works are dependents on the employees. Even if one of the employees resign the job immediately then assigned work will be not finished at the time, so delivery of the project to the clients will be delayed. Company planned to make solution for this, they want to know which employee may resign next. If they know previously, they can arrange alternative to avoid such problem. As an AI Engineer you must give Solution to this.

A) How will you achieve this in AI?

We can predict employee attrition by collect the data from company which includes historical attrition, employee tenure, workload, overtime, role changes and employee feedback. If employee feedback is positive, that employee will continue in the company. If feedback is negative, that employee may resign next.

B) Find out the 3 -Stage of Problem Identification

- **Machine Learning / Deep learning (data being in numbers)**
- **Supervised Learning – Requirement is clear and input and output data also present.**
- **Supervised learning (classification) Whether employee resign or continue in the company)**

C) Name the project: **Predict Employee Attrition**

D) Create the dummy Dataset.

Sr No	Employee Name	DOJ	Role	Overtime (Hrs)	Tenure	Employee Feedback survey	Label
1	Mr.A	10/10/23	Executive	5	3.0	Negative	Resign
2	Ms.B	12/01/24	Associate	3	2.8	Positive	Retain
3	Mr.C	05/05/23	Assistant Manager	2	3.4	positive	Retain
4	Ms.X	20/07/25	Sr. Associate	4	1.2	Negative	Resign
5	Mr.Y	09/09/23	Manager	0	3.1	positive	Retain